Example 2.11 Find the remainders when 70004 and 778 is divided by 7.

Solution Since 70000 is divisible by 7

$$70000 \equiv 0 \pmod 7$$

$$70000 + 4 \equiv 0 + 4 \pmod 7$$

$$70004 \equiv 4 \pmod 7$$

Therefore, the remainder when 70004 is divided by 7 is 4.

$\because$ 777 is divisible by 7

$$777 \equiv 0 \pmod 7$$

$$777 + 1 \equiv 0 + 1 \pmod 7$$

$$778 \equiv 1 \pmod 7$$

Therefore, the remainder when 778 is divided by 7 is 1.

Example 2.12 Determine the value of d such that $15 \equiv 3 \pmod d$.

Solution $15 \equiv 3 \pmod d$ means $15 - 3 = kd$, for some integer k.

$$12 = kd.$$

$\Rightarrow$ d divides 12.

The divisors of 12 are 1,2,3,4,6,12. But d should be larger than 3 and so the possible values for d are 4,6,12.

Example 2.13 Find the least positive value of x such that

(i) $67 + x \equiv 1 \pmod 4$ (ii) $98 \equiv (x + 4) \pmod 5$

Solution (i) $67 + x \equiv 1 \pmod 4$

$67 + x - 1 = 4n$, for some integer n

$66 + x = 4n$

$66 + x$ is a multiple of 4.

Therefore, the least positive value of x must be 2, since 68 is the nearest multiple of 4 more than 66.

(ii) $98 \equiv (x + 4) \pmod 5$

$98 - (x + 4) = 5n$, for some integer n.

$94 - x = 5n$

$94 - x$ is a multiple of 5.

Therefore, the least positive value of x must be 4

$\because$ $94 - 4 = 90$ is the nearest multiple of 5 less than 94.

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